Experiment 4

Aim :

To write and execute SQL queries using the Employee schema with INNER JOIN, LEFT JOIN, selfjoin, 3-way join, correlated subqueries, EXISTS, simulated INTERSECT, and EXCEPT, and to compare execution plans using EXPLAIN.

Description :

This experiment demonstrates different SQL techniques for retrieving and comparing data from the Employee schema. JOIN operations are used to combine related records from multiple tables. An INNER JOIN returns only matching records, while a LEFT JOIN returns all records from the left table along with matching records from the right table. A self-join is used when a table is joined with itself, such as when employees are related to their managers.

A 3-way join combines three related tables in a single query. Correlated subqueries execute with reference to values from the outer query. EXISTS checks whether a related record exists, while simulated INTERSECT and EXCEPT operations can be achieved using EXISTS or NOT EXISTS when direct set operators are unavailable. EXPLAIN is used to view the query execution plan and understand how the database engine processes a query.

Source Code :

```
-- STEP 1: CREATE DATABASE
CREATE DATABASE IF NOT EXISTS DBMS_LAB;
USE DBMS_LAB;

-- STEP 2: REMOVE OLD TABLES IF THEY EXIST
DROP TABLE IF EXISTS Project;
DROP TABLE IF EXISTS Employee;
DROP TABLE IF EXISTS Department;

-- STEP 3: CREATE DEPARTMENT TABLE
CREATE TABLE Department (
dept_id INT PRIMARY KEY,
   dept_name VARCHAR(50)
);

-- STEP 4: CREATE EMPLOYEE TABLE
CREATE TABLE Employee (
emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
dept_id INT,     manager_id
INT,
   salary DECIMAL(10,2)
);
```

```
-- STEP 5: CREATE PROJECT TABLE
CREATE TABLE Project (
project_id INT PRIMARY KEY,
project_name VARCHAR(100),
    dept_id INT
);

-- STEP 6: INSERT DATA INTO DEPARTMENT
INSERT INTO Department (dept_id, dept_name)
VALUES
(1, 'IT'),
(2, 'HR'),
(3, 'Finance'),
(4, 'Marketing');

-- STEP 7: INSERT DATA INTO EMPLOYEE

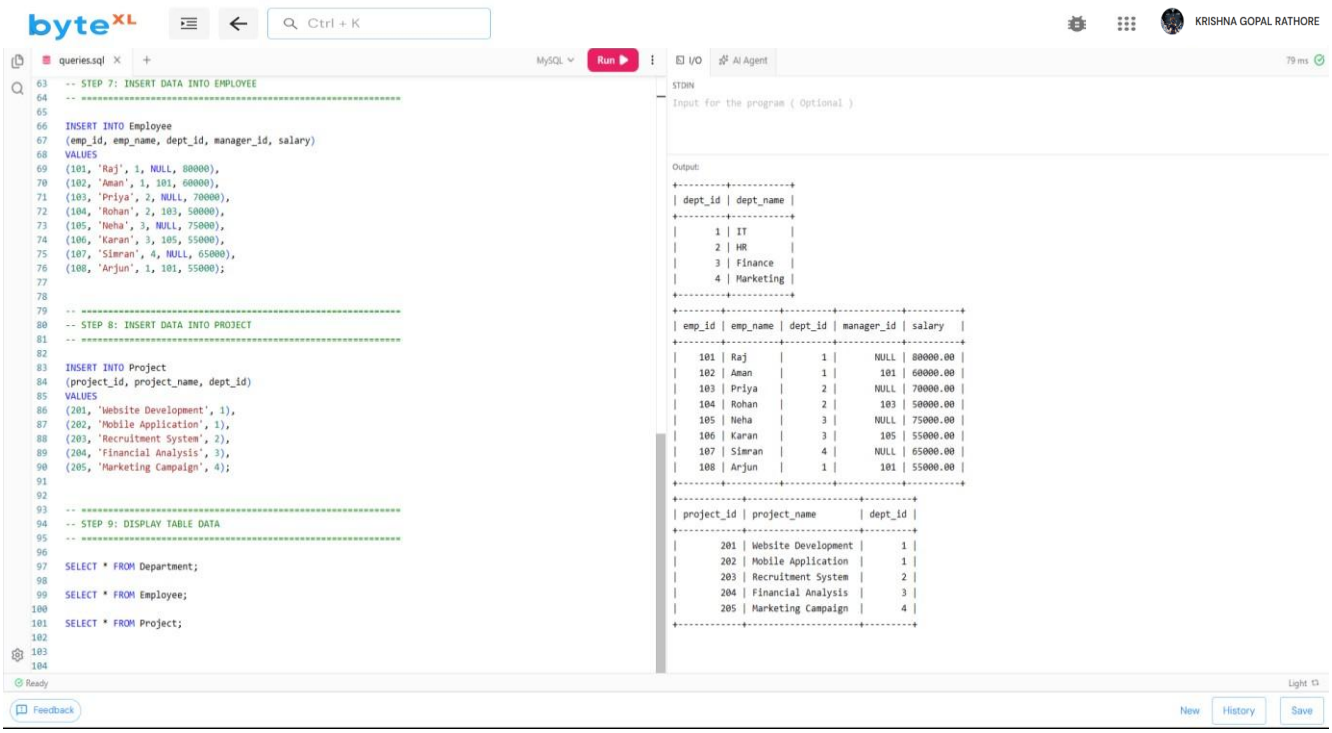
INSERT INTO Employee
(emp_id, emp_name, dept_id, manager_id, salary)
VALUES
(101, 'Raj', 1, NULL, 80000),
(102, 'Aman', 1, 101, 60000),
(103, 'Priya', 2, NULL, 70000),
(104, 'Rohan', 2, 103, 50000),
(105, 'Neha', 3, NULL, 75000),
(106, 'Karan', 3, 105, 55000),
(107, 'Simran', 4, NULL, 65000),
(108, 'Arjun', 1, 101, 55000);

-- STEP 8: INSERT DATA INTO PROJECT
INSERT INTO Project
(project_id, project_name, dept_id)
VALUES
(201, 'Website Development', 1),
(202, 'Mobile Application', 1),
(203, 'Recruitment System', 2),
(204, 'Financial Analysis', 3),
(205, 'Marketing Campaign', 4);

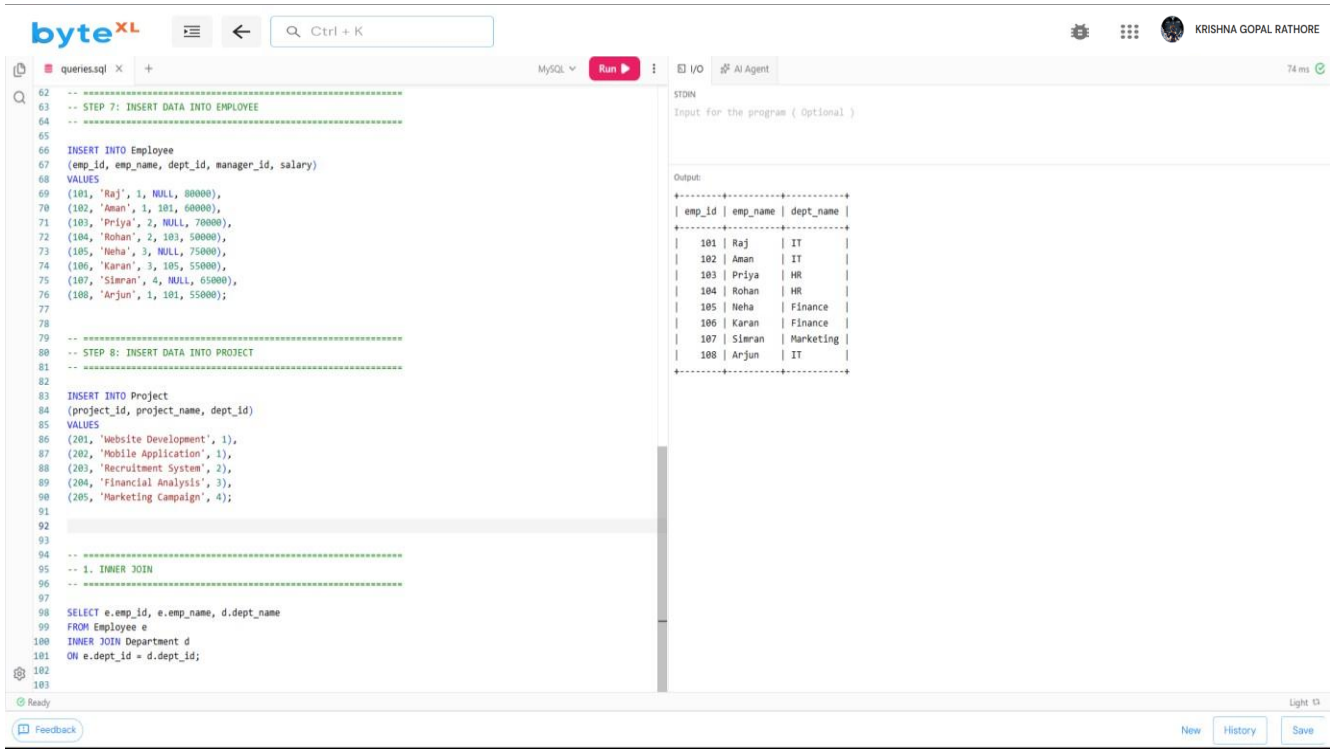
-- STEP 9: DISPLAY TABLE DATA
SELECT * FROM Department;

SELECT * FROM Employee;

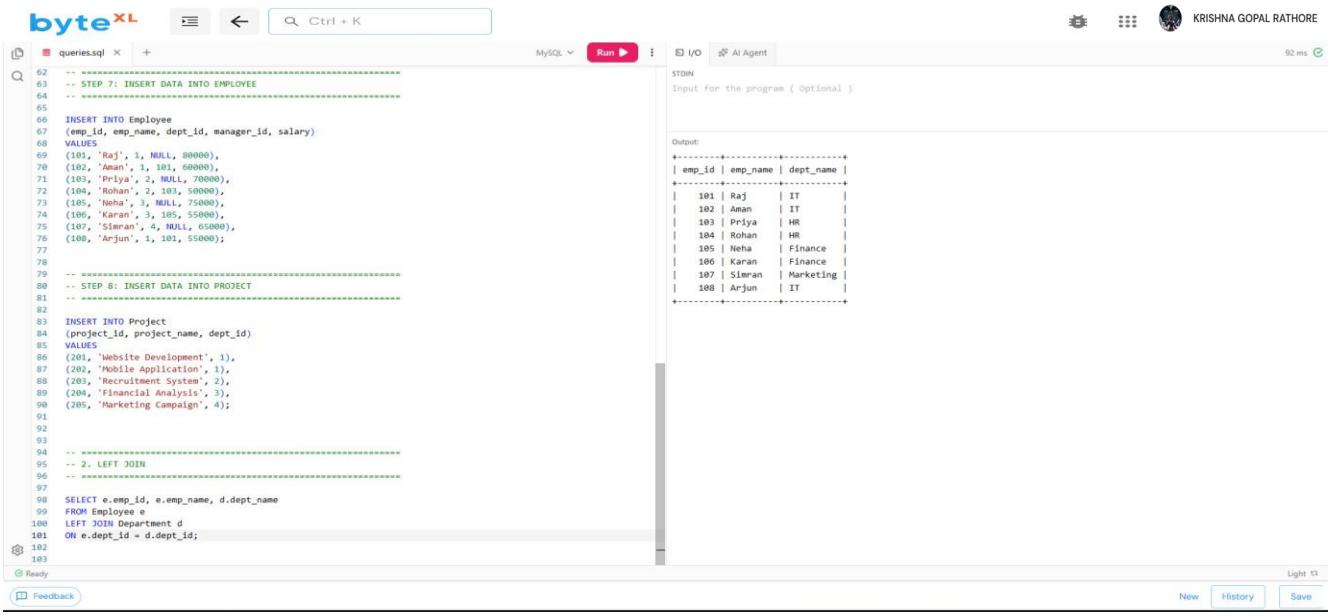
SELECT * FROM Project;
```



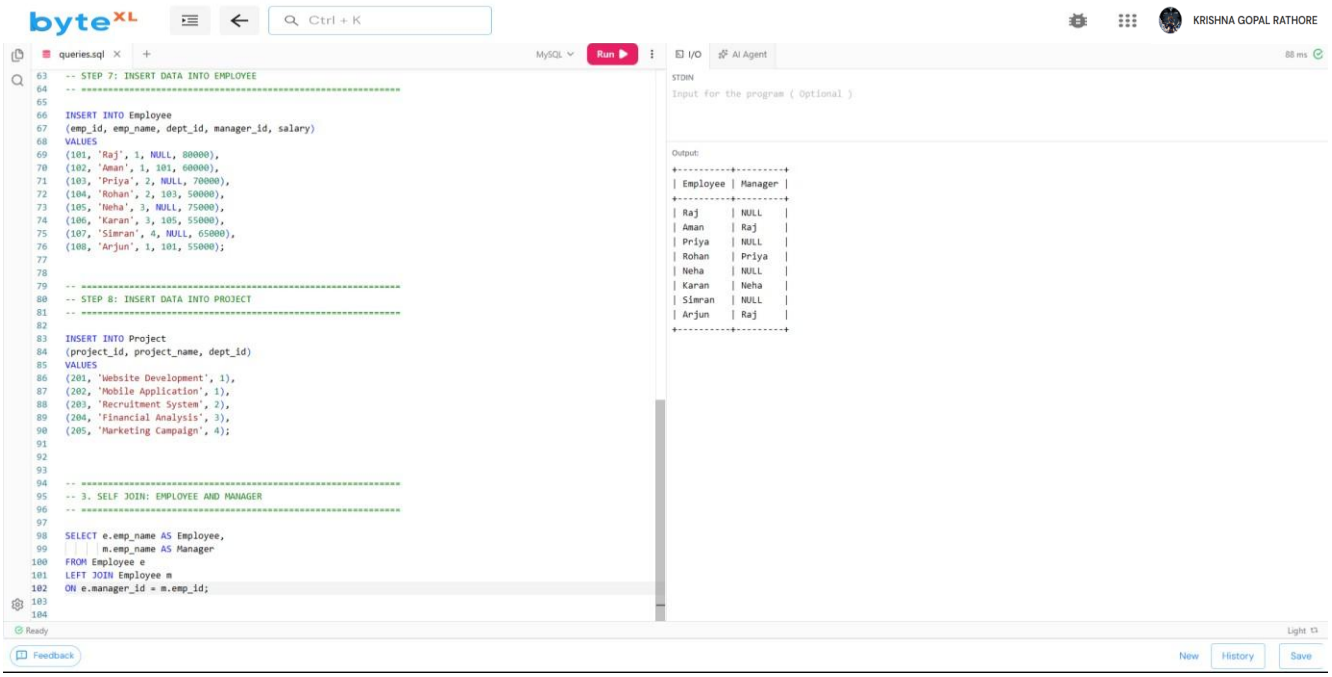

```
-- 1. INNER JOIN
SELECT e.emp_id, e.emp_name, d.dept_name
FROM Employee e
INNER JOIN Department d
ON e.dept_id = d.dept_id;
```



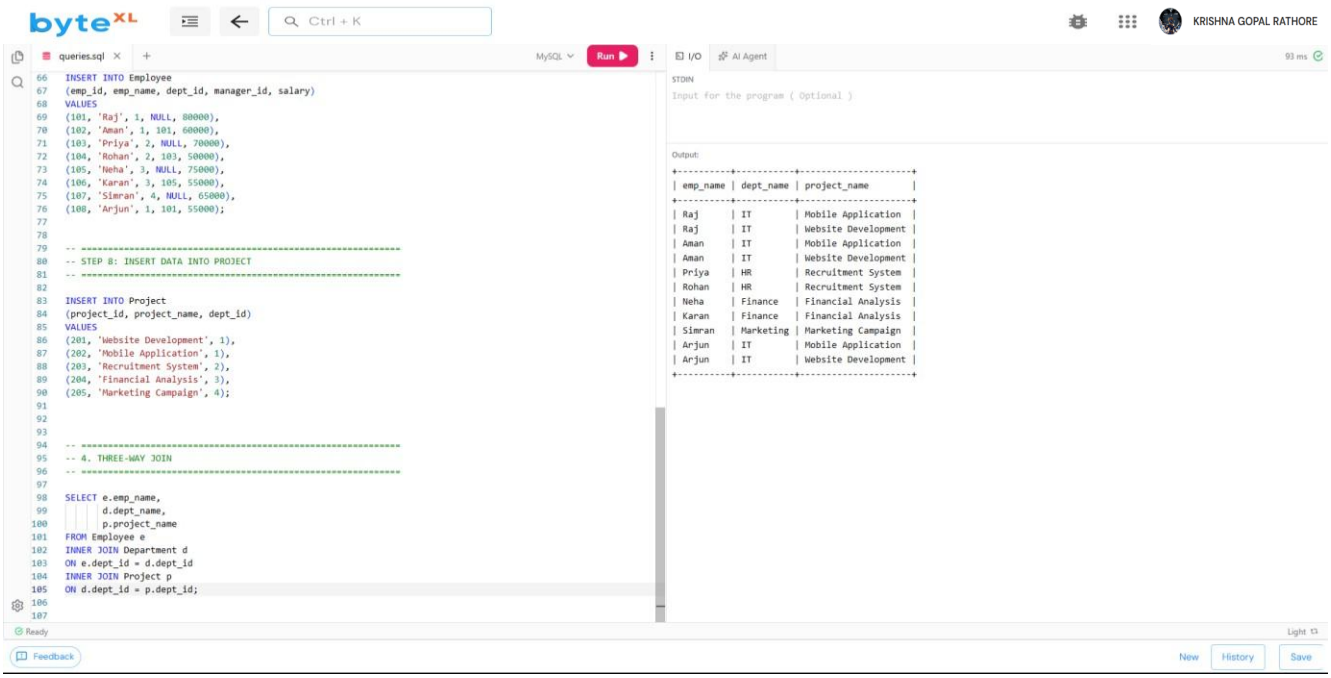
-- 2. LEFT JOIN
SELECT e.emp_id, e.emp_name, d.dept_name
FROM Employee e
LEFT JOIN Department d
ON e.dept_id = d.dept_id;



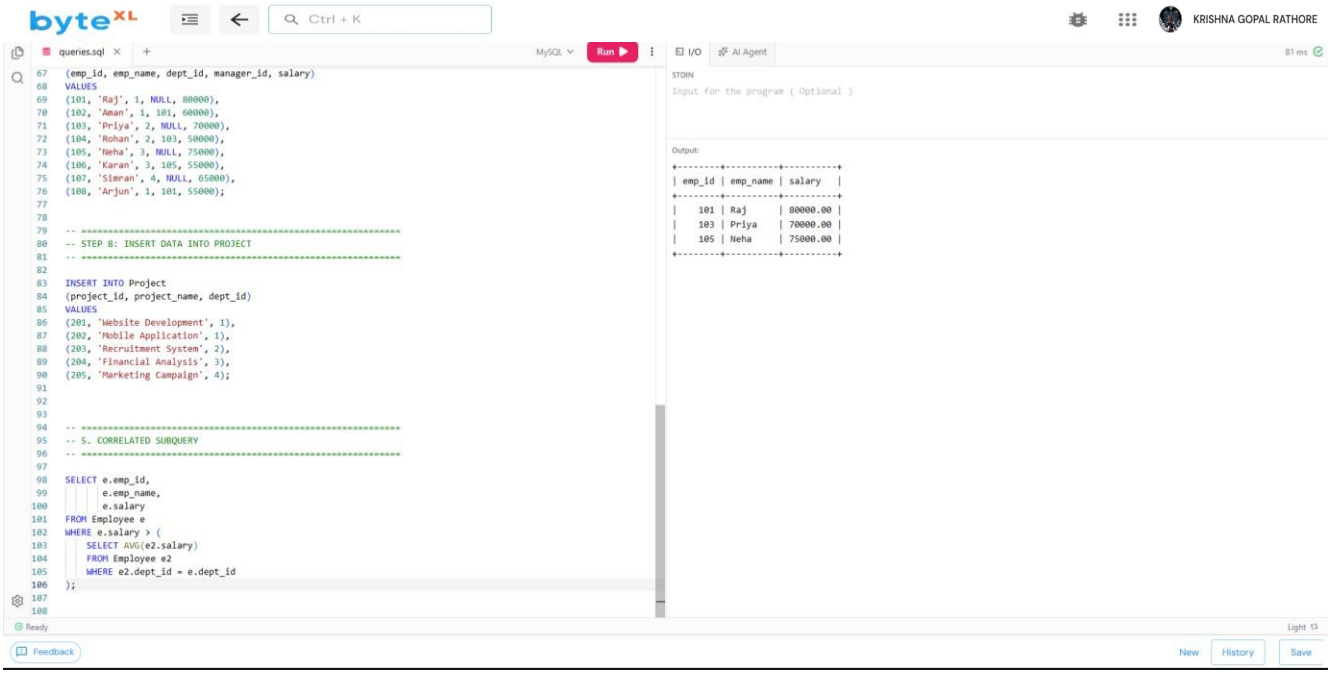
-- 3. SELF JOIN: EMPLOYEE AND MANAGER
SELECT e.emp_name AS Employee,
m.emp_name AS Manager
FROM Employee e
LEFT JOIN Employee m
ON e.manager_id = m.emp_id;



```
-- 4. THREE-WAY JOIN SELECT
e.emp_name,
      d.dept_name,
      p.project_name FROM
Employee e
INNER JOIN Department
d
ON e.dept_id = d.dept_id
INNER JOIN Project p
ON d.dept_id = p.dept_id;
```




```
-- 5. CORRELATED SUBQUERY
SELECT e.emp_id,
e.emp_name,
      e.salary
FROM Employee e
WHERE e.salary > (
   SELECT AVG(e2.salary)
   FROM Employee e2
   WHERE e2.dept_id = e.dept_id
);
```




```
(emp_id, emp_name, dept_id, manager_id, salary)
VALUES
(101, 'Raj', 1, NULL, 80000),
(102, 'Aman', 1, 101, 60000),
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-- ============================================================
-- STEP 8: INSERT DATA INTO PROJECT
-- ============================================================

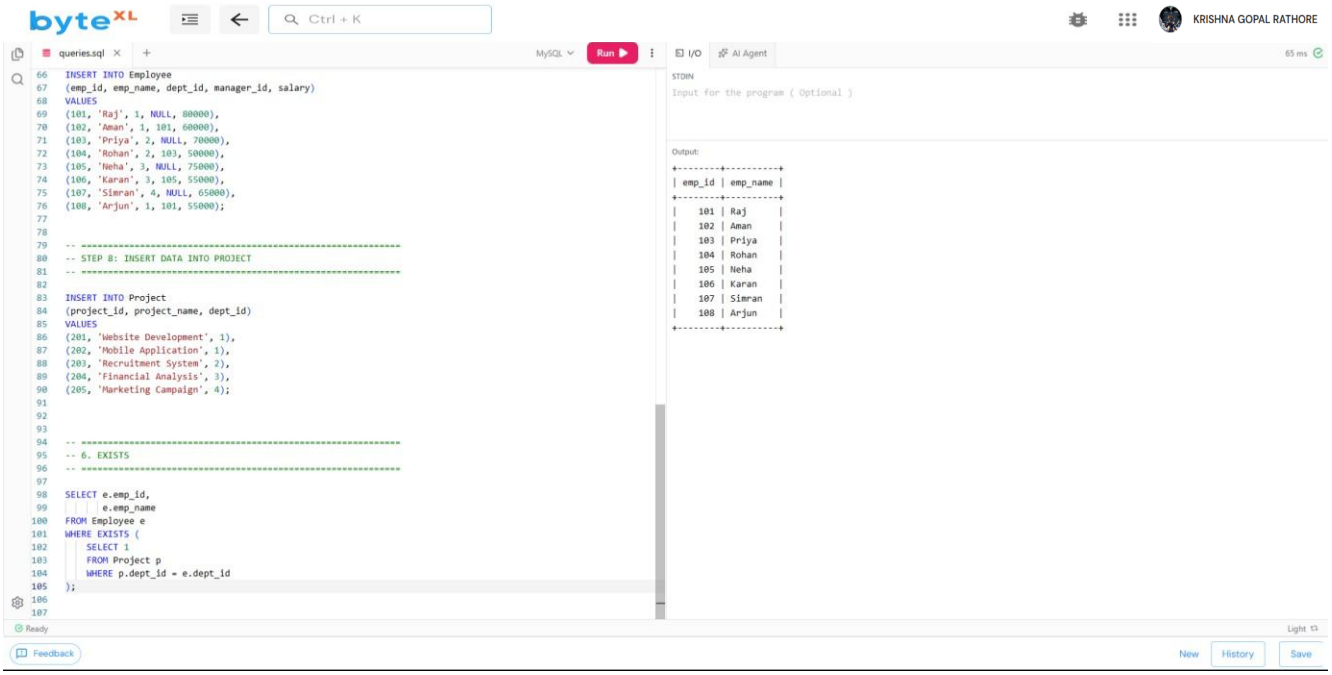
INSERT INTO Project
(project_id, project_name, dept_id)
VALUES
(201, 'Website Development', 1),
(202, 'Mobile Application', 1),
(203, 'Recruitment System', 2),
(204, 'Financial Analysis', 3),
(205, 'Marketing Campaign', 4);



-- ============================================================
-- 5. CORRELATED SUBQUERY
-- ============================================================

SELECT e.emp_id,
       e.emp_name,
       e.salary
FROM Employee e
WHERE e.salary > (
    SELECT AVG(e2.salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
);
```

```
-- 6. EXISTS SELECT
e.emp_id,
e.emp_name FROM
Employee e
WHERE EXISTS (
   SELECT 1
   FROM Project p
   WHERE p.dept_id = e.dept_id
);
```




```
INSERT INTO Employee
(emp_id, emp_name, dept_id, manager_id, salary)
VALUES
(101, 'Raj', 1, NULL, 80000),
(102, 'Aman', 1, 101, 60000),
(103, 'Priya', 2, NULL, 70000),
(104, 'Rohan', 2, 103, 50000),
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-- STEP 8: INSERT DATA INTO PROJECT
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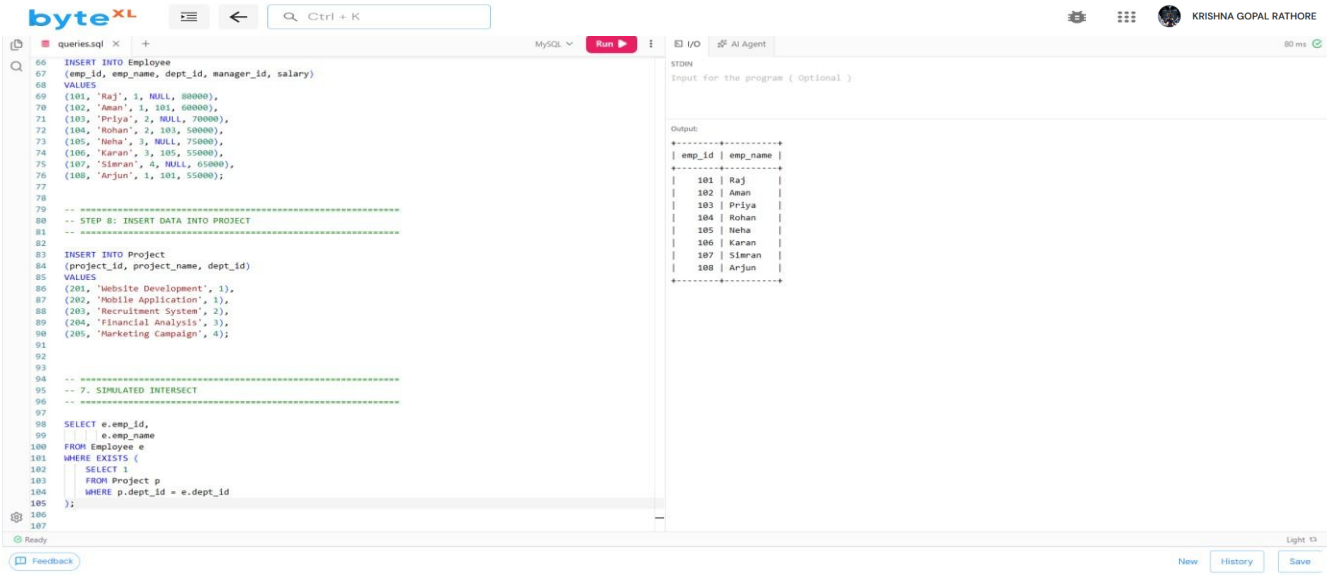
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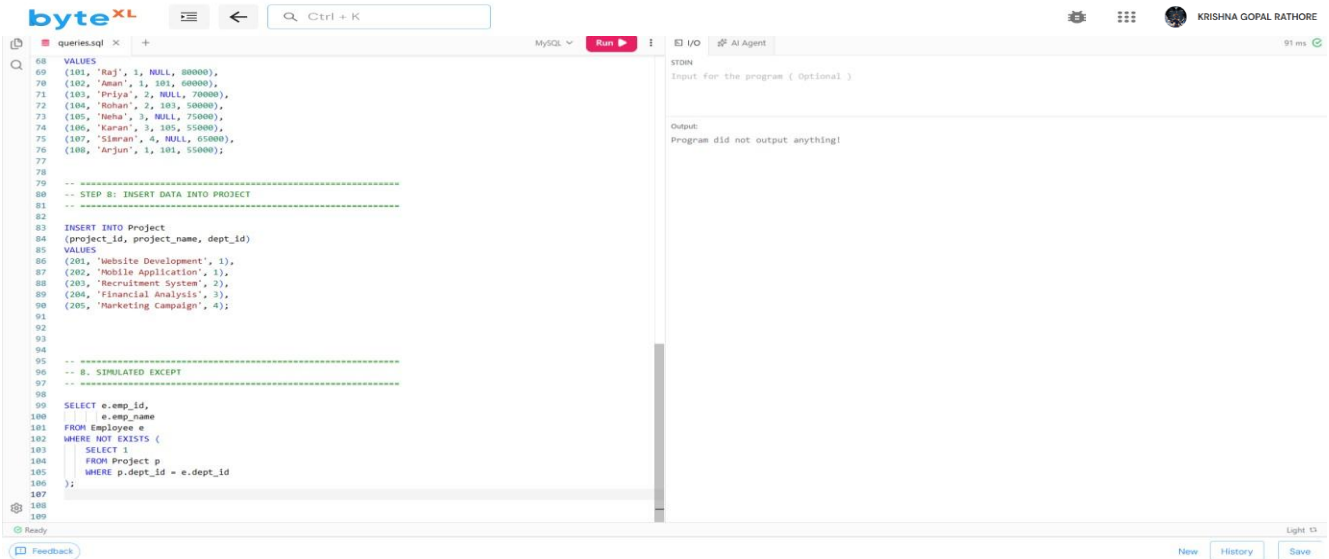
-- ============================================================
-- 6. EXISTS
-- ============================================================

SELECT e.emp_id,
       e.emp_name
FROM Employee e
WHERE EXISTS (
    SELECT 1
    FROM Project p
    WHERE p.dept_id = e.dept_id
);
```

```
-- 7. SIMULATED INTERSECT
SELECT e.emp_id,
      e.emp_name FROM
Employee e
WHERE EXISTS (
   SELECT 1
   FROM Project p
   WHERE p.dept_id = e.dept_id
);
```




```
-- 8. SIMULATED EXCEPT
SELECT e.emp_id,
e.emp_name
FROM Employee e
WHERE NOT EXISTS (
   SELECT 1
   FROM Project p
   WHERE p.dept_id = e.dept_id
);
```



```
-- 9. EXPLAIN
EXPLAIN
SELECT e.emp_id,
e.emp_name,
      d.dept_name
FROM Employee e
INNER JOIN Department d
ON e.dept_id = d.dept_id;
```

